

12. It is given that z varies directly as the cube of x and inversely as the square of y . When $x=3$ and $y=6$, $z=3$. When $x=5$ and $y=2$, $z=$

A. 5.
B. 25.
C. 125.
D. 243.

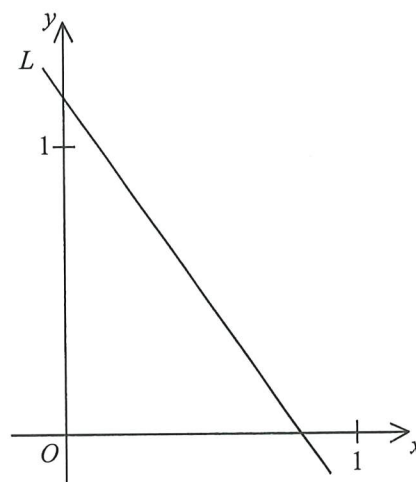
13. Let a_n be the n th term of a sequence. If $a_2=3$, $a_5=41$ and $a_{n+2}=2a_{n+1}+a_n$ for any positive integer n , then $a_6=$

A. 99.
B. 101.
C. 239.
D. 243.

14. In the figure, the equation of the straight line L is $px+qy=7$. Which of the following is/are true?

I. $p>7$
II. $q>7$
III. $q>p$

A. I only
B. II only
C. I and III only
D. II and III only



15. The perimeter of the sector OMN is 12π cm, where O is the centre of the sector OMN . If the radius of the sector OMN is 3π cm, which of the following are true?

- I. The area of the sector OMN is $9\pi^2$ cm².
- II. The perimeter of $\triangle OMN$ is less than 35 cm.
- III. The angle of the sector OMN is greater than 100° .

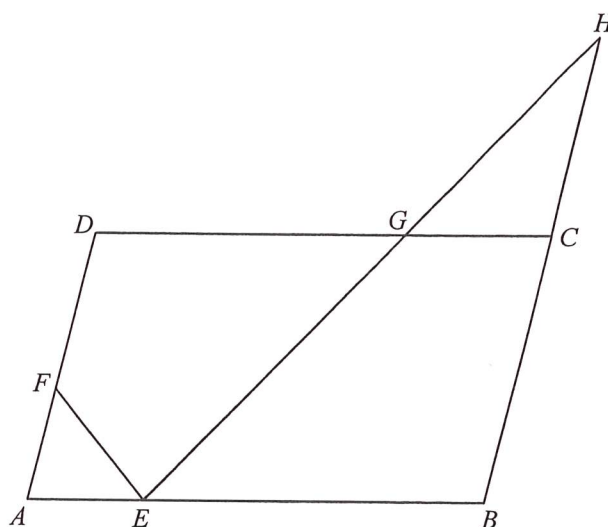
- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

16. The height and the total surface area of a solid right circular cylinder are 35 cm and 492π cm² respectively. If the radius of a solid sphere is equal to the base radius of the circular cylinder, then the volume of the sphere is

- A. 144π cm³.
- B. 288π cm³.
- C. 576π cm³.
- D. 864π cm³.

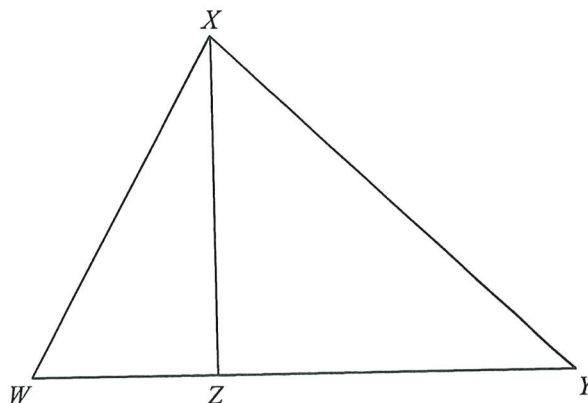
17. In the figure, $ABCD$ is a parallelogram. Let E , F and G be points lying on AB , AD and CD respectively such that $BE = 3AE$, $2DF = 3AF$ and $DG = 2CG$. It is given that BC produced and EG produced meet at the point H . If the area of $\triangle CGH$ is 16 cm², then the area of the quadrilateral $DFEG$ is

- A. 46 cm².
- B. 49 cm².
- C. 105 cm².
- D. 115 cm².



18. The figure shows the triangle WXY . Let Z be a point lying on WY such that $WY:XY = XY:YZ$. If $WX = 65$ cm, $WZ = 25$ cm and $XZ = 60$ cm, then $XY =$

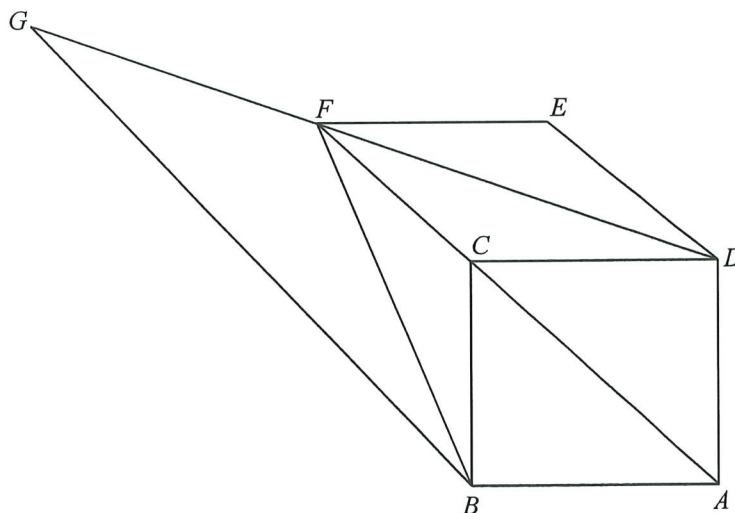
- A. 131 cm .
- B. 144 cm .
- C. 156 cm .
- D. 169 cm .



19. In the figure, $ABCD$ is a square. Let E be a point such that $CDEF$ is a rhombus, where F is a point lying on AC produced. DF is produced to the point G such that $AF \parallel BG$. Which of the following are true?

- I. $DF = FG$
- II. $\triangle BFG \sim \triangle DEF$
- III. $\angle ABG + \angle BFD = 180^\circ$

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

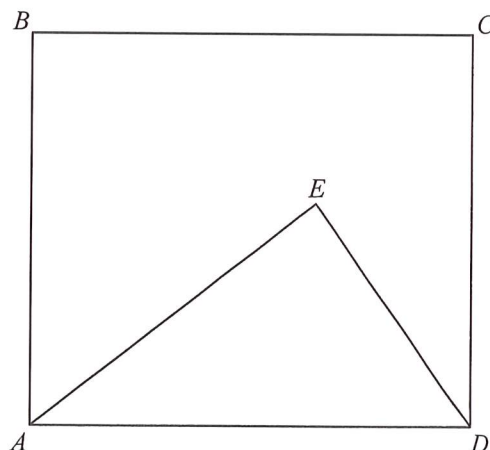


20. It is given that $PQRS$ is a trapezium with $PQ \parallel SR$. If $PS = 41$ cm, $RS = 53$ cm, $\angle PSR = 120^\circ$ and $\angle QRS = 150^\circ$, then $PQ =$

- A. 82 cm .
- B. 100 cm .
- C. 106 cm .
- D. 135 cm .

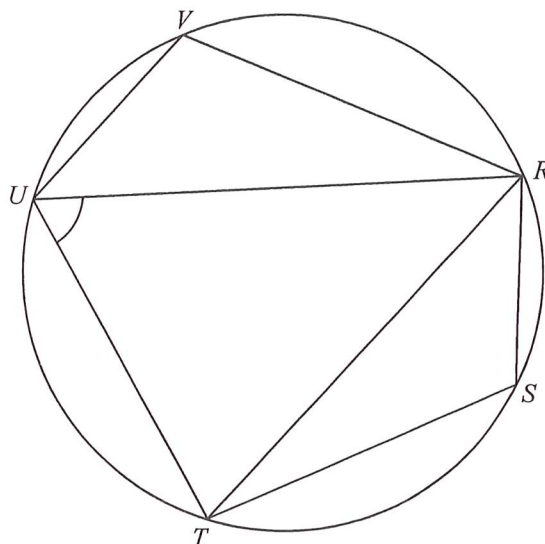
21. In the figure, $ABCD$ is a rectangle. It is given that $AE = 20$ cm and $\angle AED = 90^\circ$. If the area of $\triangle ADE$ is 150 cm^2 , then the perpendicular distance from E to CD is

- A. 8 cm .
- B. 9 cm .
- C. 12 cm .
- D. 15 cm .



22. In the figure, $RSTUV$ is a circle. It is given that $RT \parallel VU$ and RT is the angle bisector of $\angle SRU$. If $\angle RTS = \angle URV = 33^\circ$, then $\angle RUT =$

- A. 36° .
- B. 49° .
- C. 65° .
- D. 71° .



23. The figure shows the quadrilateral $ABCD$. If $\angle ABC + \angle ADC = \angle ACD = \angle BAC = 90^\circ$, which of the following must be true?

- A. $\tan \angle ACB = \frac{AB}{AD}$
- B. $\tan \angle ACB = \frac{AB}{CD}$
- C. $\tan \angle ADC = \frac{BC}{AD}$
- D. $\tan \angle ADC = \frac{BC}{CD}$

